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Russia's Arctic icebreaking capability: an assessment

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ABSTRACT

This paper explores the Russian icebreaker fleet capabilities, problems, and perspectives. It also analyses Russian civilian and military icebreakers, and their dual-use capabilities. The paper responds to a question grounded in Russia's Arctic policy: How critical is icebreaking capability to Russia's Arctic strategy? Based on the findings, the paper concludes that the influences of climate change and Russia's foreign and external security policy affect the development of its icebreaker fleet in the Arctic. In this respect, temperatures of the ice melting in the Arctic and delays in building new icebreakers, because of access to the technology caused by the sanctions, are stressed. On the other hand, Russia's confrontation with the West following the invasion of Ukraine and the sanctions over it, increases the allocation of government finance to support local shipbuilding companies. Moreover, the paper argues that Russia's economic difficulties and shipbuilding delays should enhance the dual-use capabilities of the Russian icebreakers in the coming years.

KEYWORDS

Icebreakers; Arctic; Russia's Arctic strategy; dual-use capabilities; navy; civilian icebreakers

Since Soviet times the Russian icebreaker fleet has been described, both in and outside of Russia, as one of the main factors of Russia's strength in the Arctic. Interest in Russia's icebreaker fleet has grown since the 2010s again – especially in the last few years – as Moscow's economic and security involvement in the region has increased. Moreover, Russia's 2022 invasion of Ukraine would further affect the Russian economic projects in the Arctic, including the Northern Sea Route and its icebreaker fleet. But amidst the deterioration of the Russian economy, the dual-use capabilities of the icebreakers may increase. Concerns regarding Russia's power projection into the Arctic and Trans-Atlantic, worries about Russia's purported intentions to control the Northern Sea Route, makes it necessary to develop a better understanding of the Russian icebreaker fleet's capabilities and intention.

In this respect, this paper aims to explore the Russian icebreaker fleet capabilities, problems, and perspectives on its development. The paper explores the use of icebreakers by the Russian military and also, how civilian icebreakers and infrastructure are exploited by the military. The paper identifies research questions grounded in Russia's Arctic policy: How critical is its icebreaking capability to Russia's Arctic strategy? Based on the paper's findings, it is clear that the influences of climate change and Russia's foreign and external security policy affect the development of its icebreaker fleet in the

Arctic. On the other hand, Russia's confrontation with the West, with its invasion of Ukraine and the sanctions over it, will force an increase in government finance to domestic shipbuilding companies. Furthermore, Russia's economic difficulties and shipbuilding delays may enhance the dual-use capabilities of old and new Russian icebreakers in the coming years.

In methodological terms, the paper employs qualitative methods. The research uses content analysis regarding official documents, political and military elite speeches, interviews, and reports to explain Russia's official positions and policies regarding both the military and civilian icebreaker fleets, and their use in the Arctic. Development of the icebreakers, their problems, effects of the invasion of Ukraine on the construction of icebreakers, and their dual-use capabilities are analysed in this paper. Primary and secondary sources are used to analyse developments regarding the Russian icebreakers in the Arctic. Interviews and reports from Russian and Western media are also used. The paper uses official information, reports, and interviews.

History

No nation is interested in icebreakers as much as Russia. (Vice Admiral Stepan Osipovich Makarov)¹

Since the end of the nineteenth century when the first Russian icebreaker "Yermak" was built, icebreakers have been an essential part of Russia's Arctic strategy.² In 1916, the first non-freezing port in the Arctic was founded: Romanov-on-Murman (the present name is Murmansk). By 1915, the first Russian icebreaker shipbuilding programme was created.³ The first Russian icebreakers, including Yermak, were built in English shipyards. In 1932, the USSR established the Main Directorate for the Northern Sea Route, which took over the order of new icebreakers.⁴ From 1938–1941, in Leningrad and Nikolaev, the first four Arctic icebreakers (coal-fired steam icebreakers) in the USSR were built with the lead ship "I. Stalin" (the other three were the "V. Molotov," "L. Kaganovich" and "A. Mikoyan"). The planned creation of domestic diesel-electric icebreakers in the late 1930s was postponed.⁵ They were supplied later from the USA under the Lend-Lease programme. Three icebreakers, "Southwind," "Northwind," and "Westwind" were transferred to the USSR in 1944 and early 1945.⁶ Under the names "Admiral Makarov," "Captain Belousov," and "North Pole" they operated under the flag of the USSR for several years and were returned to the United States in 1955. Thus, American vessels were the first diesel-electric icebreakers in the Soviet Union. The first Soviet diesel-electric icebreaker "Moskva" was built in Finland in 1960.⁷ Soviet-Finnish co-operation in shipbuilding began in 1932. The serial construction of icebreakers for the Soviet Union began in 1952, when a Helsinki shipyard received orders for two sea icebreakers of 10,500 hp each. The peak of icebreaker construction for the USSR occurred from 1950 to the 1980s.⁸ Since the mid-1970s, 21 icebreakers of projects 1101, 1105, 1108 and 1191 were built in Finland for the USSR, and 20 of them are still in service.⁹

After WW2, the USSR was actively engaged in the development of atomic energy. Firstly, for military purposes, to create nuclear weapons, but simultaneously the proposals on peaceful use of nuclear energy were received with the utmost interest by the Soviets. This resulted in the first nuclear-powered icebreaker in the world, the Lenin,

being launched in 1959.¹⁰ It was not only an indicator of achievement of the Soviets' science and industrial capabilities; it was also promoted by the Communist leadership as a symbol of achievement of Communism. Such approaches by the elite regarding nuclear-powered icebreakers can still be found in contemporary Russia. In the 1990s, the icebreaker fleet was strongly affected by the economic crisis and a generally diminished focus on the Arctic by Moscow. The maximum number of the Russian icebreakers serviced in the Arctic, both nuclear and diesel-electric, reached 18 in 1989, but in 2002 there were 12 operational icebreakers (7 nuclear and 5 diesel-electric icebreakers).¹¹ However, since the mid of 2000s the situation has been changing.

Contemporary Russian icebreakers

The Russian state and non-state-owned icebreaker fleets consist of more than 40 vessels, including nuclear, diesel-electric, and diesel-powered icebreakers. The Atomflot (Rosatom)¹² lead nuclear-powered icebreaker fleet¹³ currently includes: two nuclear-powered icebreakers, with two nuclear-powered icebreakers with a capacity of 81.5 thousand hp. ("Arktika" and "Sibir"), two nuclear-powered icebreakers with two-reactor nuclear power plants with a capacity of 75 thousand hp. ("Yamal," "50 Years of Victory" [50 Let Pobedy], project 10520/10521) and two icebreakers with a one-reactor installation, with a capacity of about 50 thousand hp. ("Taimyr," "Vaygach"). They are complemented by the nuclear-powered container ship "Sevmorput" (reactor capacity – 40 thousand hp). In addition, Rosatomflot operates three service vessels and a container ship "Rossita." It is also in charge of the port fleet vessels designed to serve the water area of the port of Sabetta: ice-class tugs "Pur" and "Tambey"; icebreaking tugs Yuribey and Nadym; and the "Ob" port icebreaker.¹⁴

The majority of these icebreakers were built during the time of the Soviet Union and there are now issues regarding their operational longevity. Three icebreakers of the *Arctic* class, *the Siberia*, *the Russia*, and *the Soviet Union* were decommissioned in 2010s. The service period of the reactor plants on the icebreakers currently in operation has been extended and the end of operation of nuclear-powered icebreakers will be: *50 Years of Victory* – 2039; *The Yamal* – 2028; *The Taimyr* 2027; *The Vaigach* 2026.¹⁵ The building of nuclear-powered icebreakers for Project 22220 is on-going. Their main task is to ensure year-round navigation in the Western region of the Arctic. It is planned that the icebreakers of Project 22220 will replace those of projects *Arctic* and *Taimyr*. The vessels' dual-draft concept allows for operating both in the Arctic and in the mouths of the polar rivers, particularly in the shallower areas of the Yenisei estuary and the Ob Bay area. However, in the eastern part of the Arctic they can be used only in the summer-autumn period because of more extreme weather and ice conditions in the winter-spring period. The icebreakers can overcome ice obstacles up to 3 metres thick. On 21 October 2020, a state flag raising ceremony was held on the lead universal nuclear-powered icebreaker "Arktika."¹⁶ Later it returned to the dockyard several times for substantial refits. Another icebreaker of project 22220, "Sibir" was handed over to Rosatom in December 2021¹⁷ (power – 60 MW (on shafts)); maximum icebreaking capacity – 2.9 metres; estimated service life – 40 years.¹⁸

Currently three more vessels of the Project 22220 line – Ural, Yakutia (Ob'yedinen-naya sudostroitel'naya korporatsiya, 2022) and Chukotka – are being built at the Baltic

shipyard facilities.¹⁹ In the coming years, the ships of this project will become the basis for the civilian icebreaker fleet of Russia. The icebreakers of Project 22220 are planned to have beams of 34 m, which is 4 metres wider than their “Arktika” class predecessors. It is essential for assisting large cargo ships. Moreover, this new class of icebreaker is able to combine the function of a deep-draft icebreaker – necessary for operating in the Central Arctic – and shallow-draft icebreaker for working in the mouths of the Siberian rivers. This dual-draft icebreaker takes aboard 9000 tons of ballast water and changes its draft from 10.5 to 8.5 metres by discharging ballast water. The power of this class of ships is up to 60,000 kW.²⁰ According to Deputy General Director of Glavsevmorput Vladimir Arutyunyan, the Ural will exclusively serve Rosneft’s Vostok Oil project.²¹

Another Russian prospective vessel is the “superpower” nuclear-powered icebreaker Rossiya (Project 10510 – Lider). The vessel was planned to be operational in 2027. Its icebreaking capability (maximum) will be 4 metres; displacement: draught as per CWL is about 69,700 tons.²² The Rossiya will have two RITM-400 reactors, will be 1.8 times more powerful than the RITM-200 reactors created for the icebreakers of Project 22220. The icebreaker will be capable of laying a channel about 50 metres wide. According to Rosatom, it will ensure year-round pilotage of large-tonnage vessels throughout the entire water area of the Northern Sea Route.²³ The advantage of using nuclear-powered icebreakers is their power and icebreaking capabilities. Unlike diesel icebreakers, they can operate for extended periods without refuelling. On several occasions, nuclear-powered icebreakers have remained at sea for nearly 400 days. The nuclear fleet profitability is expected to improve with the perspective increase in cargo volume along the northern sea route (NSR).²⁴

Most of the Russian diesel-electric icebreakers²⁵ were built during the Soviet Union and their numbers began to decrease due to the end of their operational periods. Currently, the Federal state unitary enterprise “Rosmorport”²⁶ operates 33 icebreakers and icebreaking tugs, including: 23 linear icebreakers, including 7 linear shallow-draught icebreakers; and 10 auxiliary and port icebreakers and icebreaking tugs.²⁷

Between 2008 and 2016, five diesel-electric icebreakers of Project 21900 with a capacity of about 18 MW were accepted for service. In 2015–2016 the construction of three 16 MW diesel-powered icebreakers was completed, 8 icebreakers of various capacities from 6.8 to 60 MW are currently under construction.²⁸ In 2020, the “Viktor Chernomyrdin” was accepted by Rosmorport for service. It is a diesel-electric linear icebreaker with a capacity of 25 MW, and according to Rosmorport, it is “the largest and most powerful non-nuclear icebreaker in the world.”²⁹ It is designed to ensure uninterrupted pilotage of ships in the Gulf of Finland, and can operate in the Arctic and Antarctic. The main operator of diesel-electric icebreakers is Rosmorport, but several vessels are owned by the Lena United River Shipping Company,³⁰ the carrier on the section of the Northern Sea Route from Khatanga Bay to Pevek, and Norilsk Nickel which has own Arctic fleet since 2006.³¹

Corporate icebreakers³²

In recent years, state and private companies involved in the exploration of natural resources in the Arctic have actively developed their own icebreaker capabilities, or invested in Rosatom’s icebreaker building projects. In 2018 and 2019, Andrey Vilkitsky

Table 1. Contemporary Russian icebreakers – nuclear powered icebreakers (The Table includes Russian operational nuclear-powered icebreakers).

Icebreaker	Year of commissioning	Building shipyard	Specs	Operator	Operationality	Ownership
"Arktika"	2020	Baltic Shipyard, Saint Petersburg, Russia	Project 22220, two nuclear-powered reactors, with a capacity of 81.5 thousand hp., icebreaking capacity – up to 3 metres	Atomflot (Rosatom)	Operational	State
"Siberia"	2021	Baltic Shipyard, Saint Petersburg, Russia	Project 22220, two nuclear-powered reactors, with a capacity of 81.5 thousand hp, icebreaking capacity – up to 3 metres.	Atomflot (Rosatom)	Operational	State
"Ural"	2022	Baltic Shipyard, Saint Petersburg, Russia	Project 22220, two nuclear-powered reactors, with a capacity of 81.5 thousand hp., icebreaking capacity – up to 3 metres.	Atomflot (Rosatom)	Operational	State
"Yamal"	1992	Russia	Project 10520, two nuclear-powered reactors, with a capacity of 75 thousand hp., icebreaking capacity – 2.25 metres.	Atomflot (Rosatom)	Operational	State
"50 Let Pobedy"	2007	Russia	Project 10521, two nuclear-powered reactors, with a capacity of 75 thousand hp., icebreaking capacity – 2.7 metres.	Atomflot (Rosatom)	Operational	State
"Taimyr"	1989	The Helsinki Shipyard, Finland	Project 10580, one nuclear-powered reactors, with a capacity of about 50 thousands hp. icebreaking capacity – 1.77 metres.	Atomflot (Rosatom)	Operational	State
"Vaygach"	1989	The Helsinki Shipyard, Finland	10580, one nuclear-powered reactor, with a capacity of about 50 thousands hp., icebreaking capacity – 1.77 metre.	Atomflot (Rosatom)	Operational	State
"Sevmorput," the nuclear-powered container ship	1988	Zaliv Shipyard, Kerch, Ukrainian SSR	Project 10081, reactor capacity – 40 thousand hp., icebreaking capacity – up to 1 metre.	Atomflot (Rosatom)	Operational	State

Source: Atomflot – rosatom.ru.

and Alexander Sannikov, diesel-electric icebreakers from Project IBSV01 (type Aker ARC 130A) started service escorting oil shipment from Arctic projects of the Gazprom Neft.³³ These new generation icebreakers operate in the area of the Arctic terminal of the Novoportovskoye field in the Gulf of Ob in the Kara Sea and ensure year-round supplies of Novy Port oil to the world market.³⁴ They are designed to perform multi-operational tasks: icebreaker escort of tankers, mooring and loading operations, towing ships, firefighting, participation in oil spill response operations, and assistance in rescue operations.³⁵

The Yamal LNG project is served by 15 gas carriers, only one of which is owned by the Russian company Sovcomflot. All 15 Arc7-classed LNG tankers are equipped dual fuel diesel-electric propulsion systems with three “Azipod” units supplied by the Swiss-based engineering company ABB.³⁶ The capacity of these gas carriers is 172.6 thousand cubic metres. They can operate at temperatures of up to minus 50 degrees and are capable of passing through ice of 2.1 m thickness. The accelerated launch of the Yamal LNG and the Arctic LNG is necessary to accelerate the expansion of the gas-carrier fleet. In this regard, Novatek placed an order for the construction of 15 gas carriers with delivery dates in 2022, 2024 and 2025.³⁷ In 2021 Nornickel³⁸ stated that it may take part in the construction of three icebreakers. In particular, the company plans to finance fully a liquefied natural gas (LNG) icebreaker, which it will need in 2024–2025. It will likely be operated by Atomflot. Nornickel, together with Rosatom, will also work out the issue of building a diesel and port icebreaker. Novatek also planned to invest in gas-powered icebreakers. According to the industry’s representatives, out of four LNG-powered icebreakers, Nornickel can potentially become an investor in one, two more by Novatek, and the financing of the fourth will be realised by coal or oil companies.³⁹ Under the agreement between Nornickel and Rosatom, the completion of another icebreaker to replace the Taimyr-type nuclear-powered icebreakers was planned. Rosatom and Nornickel also worked on the construction of port icebreakers.⁴⁰

Far Eastern Shipbuilding and Ship Repair Center JSC, Rosatom and Norilsk Nickel, within the framework of the 6th Eastern Economic Forum, signed an agreement of intent on co-operation in the design and construction of a dual-fuel, diesel-LNG icebreaker in September 2021. “The agreement is aimed at the parties’ participation in solving state problems: the implementation of investment projects in the Russian Arctic, the development of infrastructure, an increase in the cargo turnover of the Northern Sea Route, the creation of their own production of LNG icebreakers,” said the statement published by the owner of the shipbuilding company.⁴¹ The icebreaker is designed to provide pilotage for ships operating on Nornickel’s projects in the Taimyr Peninsula, as well as in the western part of the Northern Sea Route. It was planned that the icebreaker’s power-plant will operate on both diesel and gas-engine fuel, which will provide the ship with high environment-friendly engine. It was expected that the implementation of the agreement would contribute to the formation of competencies for the enterprises of the Far Eastern Shipbuilding and Ship Repair Center in the design and construction of LNG icebreakers, expanding the portfolio of shipbuilding orders and optimal utilisation of the production capacities of the Zvezda shipbuilding complex.⁴² Moreover, on the same day, 3 September 2021, Rosatom, Nornickel and Far Eastern Shipbuilding and Ship Repair Center concluded an agreement on the joint design and construction of a dual-fuel, diesel-LNG icebreaker.⁴³ Nornickel planned in

2021 to allocate about 90 billion rubles for the construction and modernisation of port facilities in the Arctic, the construction of icebreakers, including those working on LNG by 2030. The company also planned to invest 40 billion rubles in the modernisation of the Dudinsky seaport.⁴⁴

Some Russian companies operate or rent foreign ship companies' small-class icebreakers for their needs. In 2009 the multifunctional icebreaker Varandey, built at the Keppel Singmarine shipyard (Singapore), was accepted by LUKOIL to operate in the area of the Varandey oil export terminal in the Barents Sea.⁴⁵ In 2015 Keppel Singmarine transferred three ice-class vessels to the Malaysian company Bumi Armada. These ships were intended to operate on the offshore developments of the Russian LUKOIL in the northern part of the Caspian Sea.⁴⁶

Military icebreakers⁴⁷ and dual-use capabilities

The first Russian combat icebreakers appeared in the 1960s. The first of Project 97 was the Dobrynya Nikitich icebreaker, built at the Admiralty Plant in Leningrad. On 31 December 1960, it entered the auxiliary fleet of the USSR Navy.⁴⁸ Later another 30 icebreakers of this project and its modifications were built.⁴⁹ From the beginning of construction, dual-use perspectives of the Russian military icebreakers were considered. Project 97 icebreakers were exploited not only by the navy and border guards, but also by the Ministry of the Sea Transport of USSR.

In the late 1960s, a new version of the icebreaker appeared, which received project number 97P. The lead icebreaker *Ivan Susanin* was launched in 1973.⁵⁰ It differed from the previous 97th version as it lacked a nose rotor, an increased length, and a landing and take-off pad for a Ka-25PS type helicopter. The hull itself was divided by transverse bulkheads into seven compartments; unsinkability was ensured when one or two adjacent compartments were flooded. In addition, they had weapons: gun mounts AK-726 and AK-630, a rotary cannon built for self-defence. In the period from 1973 to 1981, eight ships of this project were delivered to the Navy and the border service of the USSR.⁵¹ At present, there are practically no vessels of Project 97 left. In the 1990s, most of the icebreakers of the Project 97 were dismantled. The remaining units serve in the Rosmorport and units of the Russian Navy, as well as border troops.

Currently the main task of military icebreakers is to prohibit unauthorised passage of foreign warships in the waters of the Northern Sea Route. In addition, they provide ships, as well as deployed outposts and garrisons for the fleet in the territory of the Arctic Ocean, fuel, and other necessary products. The Russian Ministry of Defence has six military bases in the High North: on the Kotelný Islands (Novosibirsk Islands), Alexandra Land (part of the Franz Josef Land archipelago), Sredniy (Severnaya Zemlya), as well as in the village of Rogachevo (Novaya Zemlya), on Cape Schmidt, and Wrangel Island (part of the Chukotka Autonomous Okrug). In addition to them, there are more than a dozen military airfields, positions for anti-aircraft missile systems, and military ports.⁵² The structure is built in such a way that the bases not only protect the territory from a possible attack through the North Pole, but also become gates to "entry and exit." The military infrastructure works year-round, but it is tied to the seasonal delivery of material and technical equipment. Therefore, icebreakers are important to the operation of military bases and infrastructure in the Arctic.

In 2017 the *Ilya Muromets*, a multi-purpose patrol icebreaker of Project 21180, was delivered to the ice-breaker group of the Russian Northern Fleet. The *Ilya Muromets* is a diesel-electric icebreaker. It is a single-deck, multifunctional vessel designed for ice-breaker support for the basing and deployment of fleet forces in ice conditions; independent pilotage of ships and vessels; towing of ships and other floating structures in icy conditions, and through clean water; extinguishing fires at emergency facilities; containment of spills and collection of oil products; helicopter reception (without basing and refuelling); transportation of containers on the open part of the upper deck, including refrigerated containers with appropriate power supply; as well as other deck and hold cargo.⁵³

Currently, two Project 23550 icebreakers – *Ivan Papanin* and *Nikolai Zubov* – are being built in the JSC “Admiralty Shipyards” in St. Petersburg.⁵⁴ They are capable of performing the tasks of a multi-purpose warships, an icebreaker and a tug. At the same time, they carry a wide range of systems, including artillery, as well as helicopters and assault boats. The ships can also carry container weapon systems, so that they can become full-fledged combat icebreakers. They can independently deliver artillery and missile strikes against sea, coastal, and even air targets. The *Ivan Papanin* has the AK-176MA automatic artillery mount on board, an icebreaker with a deck gun. The automatics of the gun and the means of supplying ammunition make it possible to fire at a rate of up to 125 rounds per minute at a distance of up to 16 km.⁵⁵

It was planned that the Project 21180M icebreaker *Evpatiy Kolovrat*, which is being built at the Almaz Shipbuilding Company in St. Petersburg, will be in the Navy’s service in 2022.⁵⁶ According to the *Izvestia*, it is being built for the Kamchatka grouping of the Pacific Fleet.⁵⁷ In December of 2022 the icebreaker started the factory sea trials in the Baltic Sea.⁵⁸

The Russian Border Guard Service has ordered two, albeit slightly modified, combat icebreakers similar to the *Ivan Papanin*. The first such icebreaker, a frontier patrol ship of the 1st rank “Purga” of ice class of Project 23550 “Ermak” was laid down in 2020 in Vyborg.⁵⁹ It will combine the capabilities of an icebreaker, a patrol cruiser and a tug. In 2020, the deadlines for the delivery of these ships to the Navy was 2024, but it is not clear whether these deadlines will be met.⁶⁰

In 2019 the chief designer of the X-35 missile at the Tactical Missile Armament Corporation stated that the Russian Navy has adopted a new concept of “open stern” which applies to Project 23550.⁶¹ According to it, highly specialised combat units for solving specific tasks are not created, but rather act as platforms. They have navigation equipment to carry out shipping tasks. And in the aft there is a free compartment where several easily removable combat modules can be placed. With this approach, it is possible to create carrier ships without tying them to the solution of a specific combat mission, and to install a combat module depending on the situation, even right at sea, while on duty. According to this concept, allegedly the aft of the ship can contain either two-tube launchers of Kalibr cruise missiles, or four-tube Uranus complexes with an X-35 missile, or two Onyxes.⁶² At present, in addition to missiles, containers can also contain various deep-sea robotic and diving complexes. Therefore, icebreakers of Project 23550, including *Ivan Papanin*, can be useful not only in surface operations, but in underwater ones as well.

However, when considering more serious support and provision of military bases, there are problems regarding the capabilities of the military icebreakers. The thickness

of the ice can – in some cases – exceed three metres, which makes it impossible to support military facilities in the event of a military conflict.⁶³

It is obvious that the military and economic activities of the region are interconnected. For instance, icebreakers of the Atomflot (the Yamal, Taimyr, Vaygach, 50 Years of Victory and Arktika) are involved in escorting ships along the Northern Sea Route, and, if necessary, providing rescuers for the military.⁶⁴ Moreover, non-military organisations are involved in cruises and even military exercises along the Northern Sea Route conducted by the Northern Fleet.⁶⁵ At various stages of such activities hydrographic vessels of the Northern Fleet, border patrol ships and icebreakers of Atomflot interact.⁶⁶ The scenarios of conducted exercises also include issues such as ensuring the safety of economic activity and resolving crisis situations or defending important industrial facilities and ports in the Arctic.⁶⁷ Carrying out such large-scale military exercises in the region is impossible without the involvement of the Atomflot, Rosmorport and other large private companies in the region. Therefore, the Ministry of Defence, the Rosatom, and other state bodies in the region need to coordinate the activities of the Navy. Public and private companies support military activities, such as providing their vessels and infrastructure for the needs of the Navy. Although the Russian government and the President's administration conduct several meetings on the Arctic problems and intend to improve interactions between state enterprises, private companies, and the military, problems amongst them appear periodically. This is mainly because, generally, in Russia, security issues outweigh economic activities and competition for control and coordination over the NSR and ports in the Russian Arctic Zone. The last issue is especially an acute problem between Atomfot and Rosmorport and, to some extent, private companies. However, currently, Atomflot has overcome its competitors on this issue.

In some cases, co-operation between military and other institutions and the emphasis on dual-use capabilities is reflected in official documents. For instance, the 2017 version of the governmental programme on the Russian Arctic Zone socioeconomic development addressed some military security-related issues by referring to the necessity of creating a dual-use transport infrastructure, including the ports of Dikson, Tiksi, Pevek, and Providence, as well as needing to maintain the technical readiness of the Northern Fleet to operate in the NSR water area. The Strategy 2020 also called for the creation of an effective Coast Guard system of the Federal Security Service.⁶⁸ It should be noted that one of the main tasks of the Russian nuclear-powered icebreaker fleet is “ensuring the safety of navigation in the Arctic, including the activities of the Navy and emergency preparedness in the Arctic region.”⁶⁹

Among other things the Russian nuclear-powered icebreakers provide for state needs, including the guaranteed performance of works and services to ensure the safety of navigation in the Arctic region, including emergency and rescue readiness, as well as the activities of the Navy.⁷⁰ Even the issue of building a new icebreaker fleet is emphasised by Atomflot leadership and experts as “one of the most important for ensuring Arctic shipping, the normal functioning of the Northern Sea Route and the implementation of control in the Arctic in the interests of ensuring the country's national security and protecting the natural environment.”⁷¹ Moreover, the Russian government developed rules for the passage of foreign warships along the NSR in 2019.⁷² Foreign military vessels should notify Russia of their plans 45 days in advance and take Russian pilots aboard. In the event of a worsening ice situation, according to the rules, only Russian

icebreakers will be authorised to provide service to foreign vessels. On 1 September 2023, changes to the new rules for navigation in the waters of the NSR came into force. In particular, a registry model is being introduced for issuing permits for navigation of ships along it. A confirmation of the issuance of a permit will be an entry made in the corresponding register maintained by the NSR General Administration, Glavsevmorput.⁷³ The primary maxim of those rules is an intention to cement Russia's control and strengthen it over the NSR.

Russia's invasion of Ukraine in 2022 and the icebreakers

Since the beginning of Russia's invasion of Ukraine, the Northern Fleet's ground forces have been used in the war.⁷⁴ Russia has few ground forces left on the Kola Peninsula in the northwest near Finland, because they were moved towards Ukraine despite Russia keeping its strategic forces in the North.⁷⁵ It suggests that the Kremlin sees the situation in the High North as stable. However, the land forces on Kola are reduced to a fifth of their original numbers before the invasion of Ukraine though its strategic power remains in the region.⁷⁶ The invasion also affects Russia's relations with other Arctic states. Recently Russia terminated its memoranda with the Nordic Council of Ministers, which includes Denmark, Iceland, Norway, Finland, and Sweden, to establish information centres in St. Petersburg and Kaliningrad.⁷⁷

Other problems and challenges endure as well. For instance, the Russian vice president of the Russian-Finnish "Helsinki Shipyard Oy" was arrested in Moscow for treason in July 2022.⁷⁸ The company was engaged in the construction of icebreakers and other vessels, mainly for Russia. The Finnish company Wärtsilä⁷⁹ announced it had completed its exit from the Russian market. The company supplied ship engines and parts to Russian shipyards, mainly for icebreakers.⁸⁰ Its departure affects current building programmes, such as that of the *Yakutia* and *Chukotka*. Instead of importing rudders and propellers, Russian-made equipment is now planned to be used, including Transmashholding's diesel engines.⁸¹

However, Russia plans to build more icebreakers and support ships, both nuclear and non-nuclear, with envisaged completions in 2027, 2030, and 2035. The plan for the development of the NSR by 2035 target, to build 2 more nuclear-powered icebreakers project 22220 by 2030, and a nuclear-technological support ship by 2027.⁸² The plan also includes construction of an additional 4 icebreakers, but does not clarify their project numbers. However, finance for 2 more previously planned "Lider"s has not yet been found. In April 2023, the *Kommersant* reported that the cost of building a lead nuclear-powered icebreaker "Russia" of project 10510 "Lider," which previously amounted to 128 billion rubles, may increase by 40–60%, and the construction period may shift. It is assumed that the icebreaker will be built at the Zvezda shipyard in 2027.⁸³

Sanctions have already started to limit building progress of icebreaker projects. For example, Norilsk Nickel and Rosneft decided not to pursue plans to build LNG-powered vessels because of sanctions. Norilsk Nickel and Rosatom decided to build a diesel-electric icebreaker, abandoning the use of a liquefied natural gas (LNG) fuel system.⁸⁴ They plan to install only Russian equipment on the icebreaker, including Transmashholding⁸⁵ diesel engines and conventional shaft lines instead of imported rudder propellers. The abandonment of the LNG fuel system is a result of Western sanctions. One of the reasons the company was keen to use LNG-powered vessels is that it is

Table 2. Contemporary Russian icebreakers – diesel-electric powered icebreakers (The Table includes Russian main operational vessels which have icebreaker capacities).

Icebreaker	Year of commissioning	Building shipyard	Specs	Operator	Operational status	Ownership
"Tor"	1964	Wärtsilä Oyj Abp	Project P-1039, power capacity 10.1 MW, icebreaking capacity – up to 1 metre	FSUE "Rosmorport" (Federal Agency of Marine and River Transport)	Operational	State
"Ivan Krusenstern"	1964	Wärtsilä Oyj Abp	Project 97A, power capacity 4500 kW, icebreaking capacity – 50 cms	FSUE "Rosmorport" (Federal Agency of Marine and River Transport)	Operational	State
"Dudinka"	1970	Wärtsilä Oyj Abp	power capacity 10688 kW, icebreaking capacity – 1 metre	Murmansk Transport Branch of PJSC MMC Norilsk Nickel	Operational	Private (Nornickel)
"Semyon Dezhnev"	1971	Leningrad Admiralty Association	Project 97A, power capacity 6118 kW, icebreaking capacity – up to 1 metre	Murmansk Transport Branch of PJSC MMC Norilsk Nickel	Operational	State
"Ivan Susanin"	1973	Leningrad Admiralty Association	Project 97P, power capacity – 3 × 1800 hp, icebreaking capacity – up to 1 metre	The Pacific Fleet	Operational	Russian Navy
"Ruslan"	1975	Leningrad Admiralty Association	Project 97P, power capacity – 2 × 2500 hp icebreaking capacity – up to 1 metre	The Northern Fleet	Operational	Russian Navy
"Yermak"	1974	Wärtsilä Oyj Abp	Project P-1039, power capacity 36 000 hp icebreaking capacity – 1.8 metres	FSUE "Rosmorport" (Federal Agency of Marine and River Transport)	Operational	State
"Admiral Makarov"	1975	Wärtsilä Oyj Abp	Project P-1039, power capacity 36 000 hp icebreaking capacity – 1.8 metres	FSUE "Rosmorport" (Federal Agency of Marine and River Transport)	Operational	State
"Krasin"	1976	Wärtsilä Oyj Abp	Project P-1039, power capacity 36 000 hp icebreaking capacity – 1.8 metres	Far-Eastern Shipping Company, FESCO	Operational	Private (FESCO)
"Kapitan M. Izmailov"	1976	Wärtsilä Oyj Abp	Project P-1039, power capacity 3920 kW icebreaking capacity – 1 metre	FSUE "Rosmorport" (Federal Agency of Marine and River Transport)	Operational	State
"Kapitan Kosolapov"	1976	Wärtsilä Oyj Abp	Project P-1039, power capacity 2.5 MW icebreaking capacity – 1 metre	FSUE "Rosmorport" (Federal Agency of Marine and River Transport)	Operational	State
"Kapitan Sorokin"	1977	Wärtsilä Oyj Abp	Project Kapitan Sorokin, power capacity 3040 kW, icebreaking capacity – 1 metre	FSUE "Rosmorport" (Federal Agency of Marine and River Transport)	Operational	State
"Kapitan Plakhin"	1977	Wärtsilä Oyj Abp	Project Kapitan Sorokin, power capacity 3 × 2100 hp, icebreaking capacity – up to 0.9 metre	FSUE "Rosmorport" (Federal Agency of Marine and River Transport)	Operational	State

(Continued)

Table 2. Continued.

Icebreaker	Year of commissioning	Building shipyard	Specs	Operator	Operational status	Ownership
"Kapitan Nikolayev"	1978	Wärtsilä Oyj Abp	Project Kapitan Sorokin, power capacity 22,000 hp icebreaking capacity – 1.8 metres	FSUE "Rosmorport" (Federal Agency of Marine and River Transport)	Operational	State
"Kapitan Chadaev"	1978	Wärtsilä Oyj Abp	Project 1105 "Kapitan Chechkin," power capacity 4650 kW icebreaking capacity – up to 1 metre	FSUE "Rosmorport" (Federal Agency of Marine and River Transport)	Operational	State
"Kapitan Dranitsyn"	1980	Wärtsilä Oyj Abp	Project Kapitan Sorokin, power capacity 22,000 hp icebreaking capacity – 1.5 metres	FSUE "Rosmorport" (Federal Agency of Marine and River Transport)	Operational	State
"Mudyug"	1982	Wärtsilä Oyj Abp	The Mudyug type icebreaker, power capacity 7.3 MW icebreaking capacity – 1 metre	FSUE "Rosmorport" (Federal Agency of Marine and River Transport)	Operational	State
"Dikson"	1983	Hietalahti shipyard	Mudyug-class icebreaker, power capacity 13,000 hp, icebreaking capacity – 0.9 metre	FSUE "Rosmorport" (Federal Agency of Marine and River Transport)	Operational	State
"Moskva"	2008	Baltic Shipyard	Project 21900, power capacity 16 MW, icebreaking capacity – 1 metre	FSUE "Rosmorport" (Federal Agency of Marine and River Transport)	Operational	State
Varandey	2008	Keppel Singmarine shipyard (Singapore)	power capacity 16.8 MW, icebreaking capacity – 1.7 metre	LUKOIL TRANS	Operational	Private (LUKOIL)
"Sankt-Peterburg"	2009	Baltic Shipyard	Project 21900, power capacity 16 MW, icebreaking capacity – 1 metre	FSUE "Rosmorport" (Federal Agency of Marine and River Transport)	Operational	State
"Baltika"	2014,	"Baltika" – JSC "United Shipbuilding Company" together with Arctech Helsinki Shipyard Inc., Projected by "Aker Arctic Technology."	multifunctional rescue vessel of project P-70202, power capacity 9180 kw, icebreaking capacity – 1.2 metres	FSUE "Rosmorport" (Federal Agency of Marine and River Transport)	Operational	State
"Vladivostok"	2015	Vyborg Shipyard PJSC (United Shipbuilding Corporation)	Project 21900M, power capacity 18 MW, icebreaking capacity – up to 1.5 metres	FSUE "Rosmorport" (Federal Agency of Marine and River Transport)	Operational	State
"Murmansk"	2016	Vyborg Shipyard PJSC in cooperation with Arctech Helsinki Shipyard	Project 21900M, power capacity 18 MW, icebreaking capacity – up to 1.5 metres	FSUE "Rosmorport" (Federal Agency of Marine and River Transport)	Operational	State
"Novorossiysk"	2016	Vyborg Shipyard PJSC	Project 21900M, power capacity 18 MW, icebreaking capacity – up to 1.5 metres	FSUE "Rosmorport" (Federal Agency of Marine and River Transport)	Operational	State
"Ilya Muromets"	2017	Admiralty Shipyard	Project 21180, power capacity 10.4 MW, icebreaking capacity – up to 0.9 metre	The Northern Fleet	Operational	Russian Navy

"Alexander Sannikov"	2018	Arctech Helsinki Shipyard, Vyborg Shipyard PJSC	Project Aker ARC 130A, IBSV01, power capacity 22 MW, icebreaking capacity – up to 2 metres	Gazprom Neft	Operational	State
"Andrey Vilkitskiy"	2018	Arctech Helsinki Shipyard, Vyborg Shipyard PJSC	Project Aker ARC 130A, IBSV01, power capacity 22 MW, icebreaking capacity – up to 2 metres	Gazprom Neft	Operational	State
"Ob"	2019	Vyborg Shipyard PJSC (United Shipbuilding Corporation)	Project the Aker ARC124 project (ice class Icebreaker7), a port icebreaker, power capacity 12 MW, icebreaking capacity – 1.5 metres	Atomflot (Rosatom)	Operational	State
"Yuribey"	2018	Craneship LLC	Ice class Arc 6, an icebreaking tug, power capacity 7 MW, icebreaking capacity – 1 metre	Atomflot (Rosatom)	Operational	State
"Nadym"	2018	Craneship LLC	Ice class Arc 6, an icebreaking tug, power capacity 6.4 MW, icebreaking capacity – 1 metre	Atomflot (Rosatom)	Operational	State
"Viktor Chernomyrdin"	2020	Admiralty Shipyard	project 22600, power capacity 25 MW, icebreaking capacity – 2 metres	FSUE "Rosmorport" (Federal Agency of Marine and River Transport)	Operational	State

Sources: Russian Maritime Register of Shipping – rs-class.org, FSUE "Rosmorport" – rosmorport.com, Federal Agency of Marine and River Transport (Rosmorrechflot) – morflot.gov.ru, United Shipbuilding Corporation – aoosk.ru, Admiralty Shipyards – admship.ru, Baltic Shipyard – bz.ru, Vyborg Shipyard – vyborgshipyard.ru, Helsinki Shipyard – helsinkishipyard.fi, Ministry of Defence of the Russian Federation – mil.ru, Portnews.ru, Moyport.info, Seanews.ru, Paluba.media, Aker Arctic – akerarctic.fi, Neftegaz.ru, LUKOIL – lukoil.com, NORNICKEL – nornickel.com, Ledokolfest.ru, Da-fish.ru.

responsible for “the worst Russian environmental disaster” in the north when 21,000 tonnes of its diesel leaked into rivers and subsoil in 2020, and after that the company announced that it will be investing heavily in higher production and improved environmental performance in 2022–2025.⁸⁶ Against this backdrop, however, Rosatom obtained financing in 2022 for the development of the eastern NSR.⁸⁷

Tasks and strategies

The tasks of the Russian nuclear-powered icebreaker fleet cover economic and strategic issues such as: ensuring year-round pilotage of ships along the Northern Sea Route in the western part of the Arctic; laying channels in drifting and fast ice to ensure the passage of ships to ports in freezing water areas; icebreaking support for the development of the Arctic oil and gas fields of the Yamal Peninsula and the shelf of the Barents and Kara Seas; provision of escort of vessels with the necessary cargoes for enterprises and settlements in the Arctic (function of the “Northern Delivery”); provision of scientific research work for solving national economic and security problems in the Arctic basin, including the polar regions of the Arctic; and ensuring safety of navigation in the Arctic, including the activities of the Navy and emergency preparedness in the region.⁸⁸

Russia’s new naval doctrine states amongst other issues the need for control over foreign military activity on the Northern Sea Route.⁸⁹ The development of the NSR is declared as one of Russia’s main maritime national interests and a strategic goal. The doctrine put “the efforts of a number of states to weaken the control of the Russian Federation over the Northern Sea Route, the build-up of foreign naval presence in the Arctic” as amongst the main challenges and threats to “national security and sustainable development of the Russian Federation associated with the World Ocean.” The doctrine declares the strategic course of the US and allies on dominance in the oceans as a main challenge and a threat to Russia’s national security in the open seas.⁹⁰

The previous 2015 Maritime doctrine also provided insight into how Moscow might engage in a maritime conflict with the West; including mobilising its civilian fishing fleet, operation of oceanographic vessels, and other ostensibly non-military vessels and installations to support a potential conflict. Michael B. Petersen stressed that the mobilisation of civilian resources for use in conflict has a long history in Russia, and this doctrine provides a clear articulation of its practice in the maritime domain. “In any case, the doctrine envisions an active role for Moscow’s improving military and its civilian resources,” he concludes.⁹¹ The new doctrine also listed measures to mobilise Russian civilian vessels, installations, and ports to support the Navy during a potential conflict.⁹² In some cases, official documents reflect co-operation amongst the military and other institutions (scientific) and companies (oil & gas) on dual-use capabilities. For instance, the 2017 governmental programme on northern socioeconomic development addressed some military security-related issues by referring to the necessity of creating a dual-use transport infrastructure, including in the ports of Dikson, Tiksi, Pevek, and Providence.⁹³

Problems and perspectives

As reflected in Russian official documents, the government targets to develop the icebreaker fleet and build new vessels despite the sanctions and challenges resulting from

the invasion of Ukraine. The design and construction of heavy icebreakers for use in the Russian Arctic represents a unique technological development where Russia considers itself the world leader, although Finnish ship designers have been at times influential partners.⁹⁴ Finnish companies are not solely involved in Russia's icebreaker building projects, as in recent years South Korean and Turkish shipbuilders also participated in such projects. The government contract for the construction of a rescue icebreaker for the Northern Sea Route was won by a Turkish company. In 2021 Kommersant reported that Kuzey Star Shipyard would become a subcontractor for the construction of two dual-fuel icebreakers (powered by diesel and LNG with a capacity of 12–14 MW) for Rosmorport at a total cost of 18.5 billion rubles.⁹⁵ The icebreakers will operate year-round in the Baltic, White and Barents Seas, as well as in the Pacific Basin, and in the seas of the Arctic Ocean and river estuaries during summer-autumn period.

In 2021 it was reported that Rosatom plans to order an additional six icebreakers to ensure the export of Russian raw materials and international transit of goods along the Northern Sea Route by 2030.⁹⁶ According to the first deputy head of the Rosatom, Kirill Komarov, two of these are nuclear-powered icebreakers of the Project 22220 series with a capacity of 60 MW, and four are powered by liquefied natural gas (LNG), each with a capacity of 40 MW. The cost of the nuclear-powered icebreaker was estimated at 60 billion rubles. The construction of new icebreakers is synchronised with the growth of cargo traffic and should be provided with appropriate agreements with shippers. These vessels will be required to replace nuclear-powered icebreakers carrying the cargo of Gazprom Neft, Norilsk Nickel and NOVATEK projects in the Gulf of Ob after 2026–2029.⁹⁷

Vyacheslav Ruksha, Deputy Director General of the State Atomic Energy Corporation Rosatom and Director of the Northern Sea Route Directorate, noted that Russia should have 12–14 icebreakers in operation to ensure year-round operation of the Northern Sea Route by 2030. “At the turn of 2030, we should have at least 7 nuclear-powered icebreakers but we need to have 12–14 icebreakers.”⁹⁸ Ruksha's statement is based on the May decree of President Vladimir Putin as it sets a goal to increase cargo turnover along the NSR to 80 million tons by 2024. This did seem to be a difficult goal before the start of the invasion, but it will be especially difficult to achieve amidst the war and Western sanctions. Moreover, in the Strategy for Developing the Russian Arctic Zone and Ensuring National Security until 2035, building additional icebreakers (one from Project 22220 and two from Lider – Project 10510) for providing annual transportation through the NSR is noted.⁹⁹ The Strategy sets a goal to increase the cargo turnover along the NSR to 90 million tons by 2030 and 130 million tons by 2035.¹⁰⁰ According to Ruksha, the cargo traffic is expected to reach 110 million tons by 2030, taking into account transit. According to Komarov, by 2030 it is planned to transport about 30 million tons of transit cargo along the Northern Sea Route. In 2020, cargo traffic along this route amounted to 33 million tons, mainly of oil, LNG, and metals from Norilsk Nickel. The implementation of the plan will largely depend on Novatek's LNG projects, which planned to increase cargo traffic to 35.5 million tons by 2024, as well as on Rosneft's Vostok Oil project, which, according to the oil company, planned add 30 million tons of freight traffic in 2024, and the Northern Star coal project by Roman Trotsenko (the plan for 2024 is about 4 million tons).¹⁰¹

In December 2021 Rosatom reported that the volume of cargo transportation through the NSR exceeded 33.5 million tons. Transit along the NSR exceeded 2 million tons. In

2020 it was about 1.3 million. The bulk of the cargo traffic was created by oil and gas investment projects: oil and oil-related products of 7.7 million tons, LNG and gas condensate of 19.6 million tons, coal of 221.5 thousand tons, and ore concentrate of 47.7 thousand tons. Other cargoes amounted to more than 4 million tons.¹⁰² Since 2016 there has been a trend of increasing maritime freight traffic (2016 – 7.3 million tons; 2017 – 9.93 million tons; 2018 – 20.2 million tons; 2019 – 31.5 million tons).¹⁰³ Despite war and sanctions, transit along the NSR increased in 2022 as well. Cargo traffic on the Northern Sea Route in 2022 reached 34.117 million tons. During 2022, cargo traffic increased primarily due to LNG transportation. In 2022 LNG transportation by the NSR increased to 1.2 million tons, container transportation grew by 226 thousand tons, and dry bulk by 109 thousand tons. The main shippers were the Russian companies: Novatek, Norilsk Nickel, and Gazpromneft.¹⁰⁴ The NSR transits in 2022 mainly were Russian outbound cargo, because after the start of the invasion of Ukraine, international shipping through the route decreased to zero. The transit of foreign cargo along the Northern Sea Route (NSR) for 2022 decreased by 10 times and amounted to 200 thousand tons, according to the Directorate of the Northern Sea Route.¹⁰⁵ However, since 2022 the NSR has been becoming critical for the Russian economy.

An earlier head of Rosatom stated that by 2030, a group of 13 icebreakers will be operational on the NSR.¹⁰⁶ In particular, nine icebreakers will operate in the western sector of the Arctic, and another four in the eastern sector. According to Rosatom's estimates, this will be enough to handle up to 150 million tons of cargo per year. "According to preliminary estimates, it is possible to double the transportations along the NSR on the horizon until 2030, and after 2034, reach the amount of freight traffic along the 'Big' NSR up to 250 million tons," said CEO Alexei Likhachev.¹⁰⁷ Moreover, Rosatom and NOVATEK planned to launch year-round navigation in eastern part of NSR in early 2024.¹⁰⁸

However, these are only plans and their occasional roles as advertisements, for both internal and external audiences, can be strikingly visible. Therefore, there are, at times, different prognoses within the same state institutions. The exact amount of sea traffic, and the number of new icebreakers will depend on several factors. There are demands for new icebreakers and their construction is planned by the state but, also with large Russian companies involved in the Arctic energy and other projects. However, Western sanctions have caused severe adjustments to the construction schedules of ships and vessels. The conditions of the Russian economy and developments in the world economy at large will influence projects in the Russian Arctic Zone, including the building of planned icebreakers. Moreover, Russia's invasion of Ukraine in 2022 has already started challenging Russia's plans. Due to the sanctions, Russian shipbuilding faced difficulties in equipping the fleet under construction with engines, propulsion, and navigation systems. Since the breakup of the Soviet Union the construction of many ships have been mostly carried out in cooperation with foreign companies. Therefore, until recently, the main suppliers of engines and diesel generators were companies from the USA, Japan, Germany, Finland, and Sweden. Although the Russian authorities consider the creation of a separate state programme "Development of engine building" with funding from the federal budget, realisation of it will require additional financial resources and time.¹⁰⁹ It is possible that Russia can import from "friendly or unfriendly" countries through using of several legal or illegal import schemes, but most probably, their quantity are not same as before 2022.

Effects of weather conditions

It is obvious that perspectives of the Russian icebreaker fleet mainly depend on the influences of climate change in the Arctic.

There will be no seasonal disappearance of ice cover in the Arctic Ocean by 2050, First Deputy Director General of the Atomflot stated: “Even in the warmest year, the Arctic seas will be ice-free only from August to October.”¹¹⁰ While Rosatom is well-known for actively advocating for the increased role of the nuclear-powered icebreaker fleet, the weather conditions alone in the Arctic waters during the winter season 2021 emphasised the necessity of icebreakers for sea navigation through the NSR. However, studies support other predictions that summers in the Arctic Ocean are likely to lose their ice cover entirely by summer 2035.¹¹¹

There are few documents available to vessels for help in navigating the NSR in various ice conditions, and information regarding permissible navigation areas and ice navigation conditions. The main official document is “Navigation rules in the water area of the Northern Sea Route.”¹¹² Data for identification of easy, moderate, or difficult ice conditions were accumulated during winter navigation of vessels by the Norilsk Nickel in the south-western part of the Kara Sea and in the Yenisei Gulf.¹¹³

Russian scholar Dumanskaya stressed that two opposite processes regarding the Russian icebreaker fleet are observed: increasing power of built and designed icebreakers on the one hand, and decreasing ice thickness in all Russian seas due to sustainable warming on the other. According to Dumanskaya, truly challenging conditions for ice navigation are occurring in the following situations: lack of icebreakers for assistance due to increased ship traffic on the route; the beam of icebreakers is insufficient to assist very large ships (e.g. a super tanker with beam of 50 m); the main icebreaker in the region is underpowered for moderate ice conditions at the sea (e.g. IB “Magadan” in the Sea of Okhotsk); icebreaking capability is decreased due to exhausted lifetime of vessels; convoy of vessels meets with a hummocked and ridged ice zone; the presence of ice class vessels not in compliance with moderate ice conditions in the sea; and a technical accident. She emphasised that the concept of easy, moderate, and difficult ice conditions corresponding with winter severity is still used by navigators to excuse problems during their winter navigation, even though ice conditions are usually not the main challenge. Obviously, when one loses the way in ice for any reason, the situation is only more complicated and makes navigating almost any ice condition difficult.¹¹⁴

As a result, Rosatom proposed to change the rules for navigation along the Northern Sea Route (NSR) in connection with the “November crisis” in 2021. It happened in a “shoulder season,” when ice conditions are especially unpredictable. Ice appeared much earlier than expected, and 20 vessels got stuck in sea ice by early November 2021. During the “November crisis” Rosatom and the ship owning companies identified important gaps in crisis response procedures. Special representative of Rosatom for the development of the Arctic, deputy chairman of the State Commission for the Arctic Vladimir Panov, said the role of long-term planning in an environment of growing cargo traffic and coordination is becoming a priority given the two operating icebreaker fleets (nuclear and diesel). It is important to increase responsibility for all participants: shippers, shipping companies, operators of nuclear and diesel icebreaker fleets, and anyone authorised sailing along the NSR.¹¹⁵

Despite fact that Rosatom is the main player in the NSR and currently it is officially an operator of the Northern Sea Route, parts of the NSR infrastructure are not controlled by Rosatom. The ports on the NSR are operated by Rosmorport, and diesel-electric icebreakers as well. Rosatom tried to take over several icebreakers from Rosmorport. This is one of the reasons why Rosatom, and its company Atomflot, actively advocated for strengthening co-ordination of navigation in the NSR in recent years. Although emphasising the need for new icebreakers and demanding the main coordination body meet Rosatom's organisational interests, its management-related problems are still acute.

Problems of management

A federal state body, the Northern Sea Route Administration, was established in Moscow by the Russian Government in 2013. The institution ensures the “safe navigation and protection of the marine environment from pollution in the water area of the Northern sea route.”¹¹⁶ Its main functions are: obtaining and issuing permission for navigation by ships through the NSR; issuing certificates of ice conventional pilotage; researching weather, ice, navigational and other conditions along the route; co-ordinating the installation of navigational aids, the harmonisation of regions to carry out hydrographic surveys operations; assistance in the organisation of search and rescue operations in the water area of the NSR; assistance in eliminating the consequences of pollution from vessels of harmful substances (i.e. sewage or garbage); rendering information services in relation to the water area of the NSR (i.e. about the organisation of navigation, requirements of safe navigation and other related issues); making recommendations about the development of routes of navigation and using icebreaking fleet in the water area of the NSR; timely data-retrieval from the Russian hydrometeorological service regarding the hydrometeorology forecast and ice analysis.¹¹⁷ The organisation, despite having the same name as the NSR Administration within the USSR Ministry of the Merchant Marine (Minmorflot), in reality received a limited mandate and is subordinated to the Agency of the Sea and River Fleet (Rosmorrechflot), under the Ministry of Transport.

The NSR's administration does not have the power to instruct other state bodies, and it does not seem to have any role in policy formation, or icebreaker-acquisition matters.¹¹⁸ According to Arild Moe and Lawson Brigham, the picture that emerges is one of two separate structures:

On the one hand, we see the formal structure with the Ministry of Transport on top and subordinate bodies to implement policy, including use of private companies or other state bodies for technical support. But we also see another structure where one of the technical bodies – Atomflot – makes the key decisions and controls the most vital information. It is the tail that wags the dog.¹¹⁹

According to the Resolution of the Government of the Russian Federation of 18 September 2020, N 1487 “On Approval of the Rules for Navigation in the Water Area of the Northern Sea Route,” organisation of navigation of ships in the water area of the Northern Sea Route is carried out by Rosatom.¹²⁰ In order to organise the navigation of ships, by the decision of Rosatom, a headquarters for the sea would be created, to provide

icebreaker escort of ships and escort of ships along the routes of navigation of ships in the waters of the NSR, as well as develop routes for navigation of ships and arranges ships of the icebreaker fleet in the water area of the NSR, taking into account the hydrometeorological, ice and navigation conditions in the specified water area.

The Resolution stated that there is a permitting procedure for the navigation of ships along the NSR and permits are issued by the Federal Agency for Sea and River Transport, or an organisation subordinate to it in agreement with the headquarters.¹²¹

The document stressed that icebreaker escort of vessels, the navigation of which in the waters of the NSR, “according to the admission criteria, must be carried out under icebreaker escort, is performed by icebreakers entitled to sail under the State flag of the Russian Federation.” According to the resolution, icebreaker assistance is carried out in accordance with the rules for icebreaker assistance of ships in the waters of the NSR,

approved by the Ministry of Transport of the Russian Federation in agreement with the State Corporation Rosatom, and includes the navigation of a vessel (convoy of ships) with the help of an icebreaker (icebreakers), as well as the activities of the icebreaker fleet, ensuring the specified navigation, including the formation of a convoy of ships and the placement of ships in the convoy to follow icebreakers, preliminary laying of a channel in ice, towing of ships in ice conditions, ice reconnaissance by an icebreaker, ensuring safe anchorage (drift) of ships in ice while waiting for an improvement in ice conditions.¹²²

Although Rosatom succeeded to obtain the rights over the infrastructure operator of the NSR, there was still a diarchy represented by the Northern Sea Route Administration/the Ministry of Transport and the headquarters of Marine Operations of Atomflot/Rosatom until August 2022. On 1 August 2022, the Federal State Budgetary Institution “Main Directorate of the Northern Sea Route” began work as part of the Rosatom.¹²³ It included the headquarters of marine operations. It means since then the management of navigation along the NSR has been controlled by one body.

In addition to icebreakers and navigation-related problems, perspectives of the NSR and generally the Russian Arctic region require the development of coastal infrastructure, in particular the eastern Arctic. Building new icebreakers and developing port infrastructure will allow the use of ships more than 40 m wide with the displacement of more than 100 thousand tons for cargo transportation.¹²⁴ However, even before the start of the Russia’s invasion 2022 of Ukraine, in 2021, port infrastructure investors have faced challenges associated with the crisis caused by the coronavirus and unpredictable changes of logistic chains.¹²⁵

Conclusion

The transport crisis in the NSR in November and December 2021 exposed several long-standing problems, in particular problems with transportation management, the NSR and the state institutions of the Russian Arctic, and most importantly, with co-ordination of plans and activities of all the Russian Arctic players. Moscow tried to solve its main problems of co-ordination by transferring the main responsibilities of the NSR management to Rosatom in 2022. Russia’s invasion of Ukraine has affected all of Russia’s Arctic projects, energy extraction, the development of the NSR, and shipbuilding. Because the sanctions followed the invasion limit Russia’s state and private companies’ accesses to the Western investments and technology, which are important for the realisation of Russian Arctic projects. Even China and other potential investors also have concerns trying to involve in Russia’s projects over the sanctions. However, Moscow intends to meet the needs of the Russian projects in the Arctic with the involvement of non-Western actors, such as Asian and Middle Eastern states.

Based on these developments, this paper argues that the influences of climate change and Russia's foreign and external security policy have affected the development of its icebreaker fleet in the Arctic. On the other hand, Moscow's confrontation with the West amidst the invasion of Ukraine and the sanctions over it increased the allocation of the government's finance to local shipbuilding companies. Moreover, Russia was already heads and shoulders ahead of the rest in icebreaker capability, and the fears of sanctions/Western curtailment of investment in the industry, have not fully materialised yet. In such circumstances, however, Russia's economic difficulties and shipbuilding delays should enhance the dual-use capabilities of old and new Russian icebreakers in the coming years. Developments of the Russian icebreaker fleet will mainly depend on the weather of the Arctic, developments of the NSR, general economic and financial trends in the world, and Moscow's international behaviour has brought about Western economic sanctions in recent years.

Notes

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